

Basketball 3D Game — Unity Asset by makaka.org

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[Basketball \(AR and Non-AR\)](#) — cross-platform Unity Asset in the sport arcade genre with realistic physics of basketball net & ring, ring growing & hoop movement, and advanced scoring & audio systems.

Note: The asset is downloaded into your project as a Template (Complete Project) and has a free external dependency. [Read the "Getting Started" Tutorial](#) to Import it correctly.

Check out the [Support section](#) if you need help.

During an encounter with the Basketball Hoop, a player may throw the ball by:

- ★ clicking/tapping any point on the screen (by default);
- ★ flicking/swiping the object from the bottom of the screen up toward the target.

Both modes form different speeds and throw directions based on the last position of the finger or mouse cursor, what makes the game even more interesting. Throw Diligently and Try to Get the Best Score without touching the Ground!

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Game Modes: AR and Non-AR

You can choose Game Mode on the Demo Scene before the start:

1

Non-AR Mode (for [all tested platforms](#)) with the next Camera Control Modes:

- ★ *Desktop*: Keyboard/Mouse.
- ★ *Mobile*: [Sensor Camera \(docs\)](#).

2

AR Mode (if it's available, the user can see a button to activate it) with the next features using the [AR Foundation \(ARCore, ARKit\)](#) — Augmented Reality engine:

- ★ Real Plane Detection (includes tutorial, visual plane confirmation/resetting, placing the game around the player).
- ★ [AR Safe Zone](#). The player needs to stay in the Safe Zone to avoid accidents during the game and continue the game itself! The player will be notified when leaving the Safe Zone.
- ★ AR Onboarding UX with Transparent Video Manuals & AR States.
- ★ [AR Shadows & Light Estimation](#): estimates light data in the real world and applies it to game space.



#AR #Basketball 🏀 #2 — #Unity Asset — ARKit, AR
Makaka Games



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Package Contains

- ★ Unified Demo Scene for Both Modes (AR and Non-AR).
- ★ [Sensor Camera \(docs\)](#).
- ★ [Throw Control \(docs\)](#).
- ★ Menu Scene.
- ★ Permission Scene with Camera Permission Request using free [Native Camera](#).
- ★ Loading Screen to switch scenes seamlessly.

[Check the Map of Unity Assets](#) to choose the product that best suits your needs.



#Basketball Game 3D 🏀 — #Unity Asset — #Gam
Makaka Games



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Related Unity Assets

[Football \(docs\)](#).



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Features of Basketball Game 3D

Throw like a Pro with your amazing Game or App:

- ★ 2 Bonuses: Ring Growing (Big Ring), Hoop Movement.
- ★ 2 Throwing Modes with [Throw Control \(docs\)](#): "Click" (Easy) & "Flick/Swipe" (Hard).
- ★ 100+ Customizable Parameters (Points, Throwing, Bonuses, Sounds, Delays, etc.).
- ★ Super Realistic Physics of a Basketball Net & Ring:
Bank Shot, Toilet Bowl Shot, Rimshot, Net Shot, Clear Shot...
- ★ Dissolving (Fading) VFX for All Game Objects.
- ★ Dynamic Sound System: play Sounds based on speed, pitch, and volume factors of the Ball when it collides with Ring, Basketball Net, Backboard, Floor, Pole.
- ★ Directional Arrow that points to the Current Target.
- ★ Advanced Scoring System with Combos & Accounting Distance to Basket.
- ★ UI Animations and Sounds: Clear Ball, Best Score, Current Score, Fail.
- ★ Simulation of Player Breathing (Desktop).

Bonuses

You can get a bonus with 1 scored ball or with a combo. Balls Count in Combo can be set in Unity Editor.

Ring Growing (Big Ring)

Ring Growing happens when the player scores 1 Clear Ball (without touching of Ring). When the Limit of Goals with Big Ring will be reached then the basket will be returned to the Normal Size.

Rare Clear Shot 🏀 3D #Basketball Game 🏀 #Uni
Makaka Games



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Hoop Movement

Hoop Movement happens when the player scores 2 Normal Balls (not clear) and shows VFX with Explosion & Hoop Dissolving (Meshes & Shadows).

Movement contains 2 independent parts:

- ★ Rotation Around Camera (Player) by Target Angles.
- ★ Changing of local Z Position by target step.



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Basketball Net Demo

Video Demo below of Basketball Net is just an example of an Interactive and fully Touchable Basketball Net with 2 holes, using the [Cloth component](#) and the [Skinned Mesh Renderer](#) to provide a physics-based solution for simulating fabric.

At the same time, the complete [AR Basketball project](#) provides an [AR Safe Zone](#) and the player will be given a warning about leaving the safe zone, the screen will be darkened and throws will be blocked.



Cloth in #Unity (Basketball Net) #GameDev #AR

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User Input

Menu Scene

Action	Desktop	Mobile
Play	LMB / Space	Tap
Reset Data	LMB / R	Tap
Quit	LMB / Escape	Tap

Demo Scene

Action	Desktop	Mobile
Throw	LMB	Tap
Rotate Camera	RMB / WASD / Arrow Keys	Non-AR: Motion Sensor (Gyroscope or Accelerometer) via <u>Sensor Camera</u>

Action	Desktop	Mobile
		AR: AR Foundation scripts (provide the same experience as in Non-AR)

Limitations

Learn Limitations separately for Each Asset included in the package.

Tutorial



*This tutorial is relevant for Basketball Game 6.0.
Tutorial for the previous version can be found
only in the asset folder.*

Getting Started with Basketball Game 3D

Folders & Files in the package by default:

- ★ Makaka Games;
- ★ XR.

Steps



*If you have any issues with the first launch then
just Reach Support and Get Help.*

- 1 Create a New Unity Project with [Unity 6000.3.11](#) & the "Universal 3D" Template.
- 2 File > Build Profiles > Platforms > Windows, Mac, Android, iOS, Web > Switch Platform.
- 3 Next Packages must be manually installed with [Git & Unity Package Manager](#). I recommend using indicated versions to get a guaranteed experience:

★ [Native Camera 1.5.0](#).

Hash value for

installing: [72f5ebba26c219b4c4cda202bff2f59bfdaa47f8](#)

- 4 Download and import [Basketball Game](#) into Unity.
 - 1 Go through the Warning Windows:
 - 1 Click "Import" to overwrite the Project Settings with predefined ones.
 - 2 Click "Install/Upgrade" for Unity Package Manager dependencies.
 - 2 "Import Unity Package" window:
 - 1 Click "Next".
 - 2 Click "Import".

5

Next Packages are provided with [Unity Package Manager](#), and they are already installed for this Asset by default. If packages are missing (Warning Window did not appear) then install them again with Package Manager:

1

[uGUI 2.0.0:](#)

★ **Always Required:** Window > TextMeshPro > Import TMP Essential Resources.

2

[AR Foundation 6.3.3.](#)

3

[Apple ARKit XR Plugin 6.3.3.](#)

4

[Google ARCore XR Plugin 6.3.3.](#)

5

[Input System 1.19.0.](#)

6

Reopen Unity Project.

7

Open Scene: Makaka Games > Basketball > Scenes > Demo.

8

[Test in the Unity Editor or Build.](#)



Each Unity Asset included in [Basketball Game](#) has its own documentation in the target folder or [on the website \(the latest docs\)](#).

"\$" GameObject

Some Scenes have \$ Game Object, which has the main control script for the scene.

Various modules interact with each other with the \$.

How to Set Balls Count & Throwing Mode?

All the information about throwing you can find in the ["Throw Control" Documentation](#).

3rd Party 3D Basketball Environment (Option)

[Basketball Game 3D \(Unity Asset\)](#) is not related to the next optional assets, and it doesn't include them and support is provided only by their publishers. The next assets are recommendations as additional stuff for inspiration and application in your unique game:

- ★ [HQ Abandoned School \(Modular\)](#): 2 basketball areas (street playground & school gym).
- ★ [Basketball Arena](#): Low Poly Basketball Stadium, Set to NBA dimensions, perfect for basketball game development, includes basketball hoops and basketballs, optimized for realtime rendering.
- ★ [School Gym](#): common school gym with basketball stuff.

Testing and Building

[All tested platforms](#) support Non-AR Mode. If AR Mode is available, the user can see a button to activate AR Mode on the Demo Scene before the start.

iOS, Android: AR & Non-AR

AR Mode can be tested in the Unity Editor with [AR Foundation > XR Simulation](#).

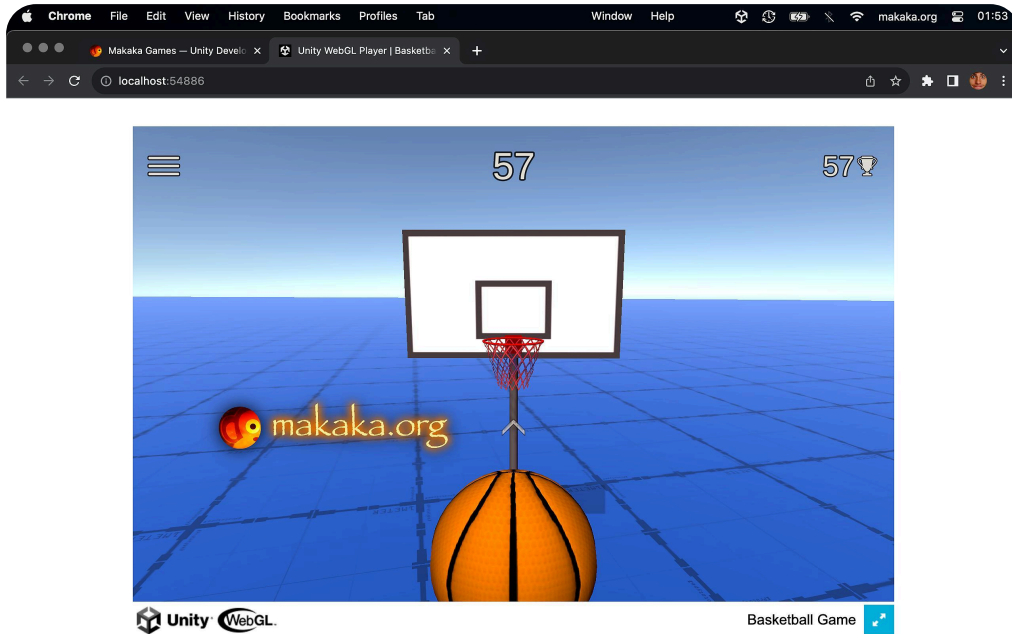
macOS, Windows: Non-AR

You can test immediately in the Unity Editor from the box, but to Make a Build, you need to Uncheck the Next Flags in the Edit > Project Settings > XR Plug-in Management > Windows/Mac settings:

- ★ *Initialize XR on Startup*.
-

WebGL: Non-AR

Learn the Article called [WebGL and Unity](#) about Building and Testing Unity games and apps for WebGL.



[Basketball Game 3D](#) on WebGL (Desktop) in Google Chrome.

Tested with Platforms

1

Mobile Platforms:

- ★ **AR & Non-AR: iOS** on iPhone 15;
- ★ **AR & Non-AR: Android** on Samsung Galaxy A71;
- ★ **Non-AR: WebGL** in Google Chrome.

2

Desktop Platforms (*Non-AR*):

- ★ **Windows**;
- ★ **macOS**;
- ★ **WebGL** in Google Chrome.

Support

First, [read the latest docs online](#).

If it didn't help, get the support:

★ [Unity Discussions.](#)

★ [Email.](#)

Changelog

Check the current version of [Basketball Game 3D on the Asset Store.](#)

The latest versions will be added as soon as possible.

6.0 (Unity 6.3):

Features:

- ★ **New Menu Scene** with the Next Buttons (Keyboard can be used in addition to the Mouse):
 - ★ Play (Space key),
 - ★ Reset Data (R key),
 - ★ Quit (Escape key).

Improvements:

- ★ [Unity 6000.3.11](#) with Modern Standards of Project Settings (based on New Unity Project).
- ★ [URP \(Universal Render Pipeline\)](#) to improve performance (instead of the Built-in Render Pipeline).
- ★ [Input System](#) (New Standard in Unity) for All Project-related Scripts: *instead of [Input Manager.](#)*
- ★ [AR Foundation 6.3.3.](#)
- ★ [Sensor Camera 3.2.](#)
- ★ [Throw Control 7.2.](#)
- ★ Android: Vulkan Graphics API first (and Linear Color Space).
- ★ Remove Unused Tags & Layers.
- ★ Namespaces for All Scripts.

5.0 (Merging this Unity Asset with AR version into 1 Scene: AR and Non-AR):

Features:

- ★ **AR Mode for iOS & Android:** [AR Foundation \(ARCore, ARKit\)](#) + Plane Detection (includes tutorial, visual plane confirmation/resetting, placing the game around the player):
 - ★ [AR Safe Zone](#). The player needs to stay in the Safe Zone to avoid accidents during the game and continue the game itself! The player will be notified when leaving the Safe Zone.
 - ★ AR Onboarding UX with Transparent Video Manuals & AR States.
 - ★ [AR Shadows & Light Estimation](#): estimates light data in the real world and applies it to game space.
 - ★ Editor Testing with [AR Foundation > XR Simulation](#).
- ★ **"Select Mode" Panel** (on the Demo Scene before the start, you can choose Game Mode):
 - ★ **AR Mode** (if it's available, user can see a button to activate it),
 - ★ **Non-AR Mode** ([all tested platforms](#) support it).
- ★ Using of [Native Camera](#) to check Camera Permission before the main scene.

Improvements:

- ★ [Unity 2022.3.20](#).
- ★ [Throw Control 6.0 \(docs\)](#).
- ★ [Sensor Camera 2.1 \(docs\)](#).
- ★ Caching of Game Objects to avoid freezes at the very 1st start of the game.

4.1:

Improvements:

- ★ [Unity 2022.3.18](#).

Fixes:

- ★ Lighting in Graphics Settings for iOS & WebGL.

4.0 (Mobile WebGL Support):

Features:

- ★ WebGL Support via HTTPS for Mobiles: Android.
- ★ Check information about iOS in the [Limitations section of Sensor Camera](#).

Improvements:

- ★ [Unity 2022.3.7](#).
- ★ [Sensor Camera 2.0 \(docs\)](#).

Learn the DevLog in the Video Format below.



How to Fix Bugs in Unity – Rigidbody & Cloth – T

Makaka Games



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3.0:

Features:

- ★ Support for Mobile Platforms (**iOS & Android**) using [Sensor Camera 1.0 \(docs\)](#):
without Simulation of Player Breathing.

2.4:

Improvements:

★ [Unity 2021.3.21.](#)

★ [Throw Object 3D 5.2 \(docs\).](#)

Fixes:

★ Camera Rotation Sensitivity with Mouse for Different FPS and Platforms.

2.3:

Improvements:

★ [Unity 2021.3.4;](#)

★ [Throw Object 3D 5.1 \(docs\).](#)

2.2:

Improvements:

★ [Unity 2021.2.16.](#)

★ Modern Standards of Project Settings (based on New Unity Project).

★ [Throw Object 3D 5.0 \(docs\).](#)

★ Unified UI Layout to match different resolutions.

★ Soaring Style for HUD UI with Black Outline: no black panel behind anymore.

★ Loading Animation for Start Button.

2.1:

★ New Platform: WebGL Support (Tested in Google Chrome).

★ [Unity 2021.1.12.](#)

★ [Throw Object 3D 4.3 \(docs\).](#)

2.0 (Completely Rewritten Project & New Features):

- ★ Throw Object 3D 4.0 (docs);
- ★ TextMesh Pro;
- ★ Hoop Movement;
- ★ Dissolving (Fading) VFX;
- ★ Shadow for Basketball Net;
- ★ New Camera Control Mode: Right Mouse Button;
- ★ Improved Basketball Physics Material.